Module 05: Time Series Analysis and Predictive Modeling

Mohan Babu University - Data Science Digital Library
Comprehensive Study Curriculum - 10 Detailed Chapters

Course Overview:
This module provides an exhaustive study of time series forecasting and predictive modeling. Students will explore statistical decomposition, stochastic models, parameter estimation, feature extraction from temporal signals, and modern machine learning/deep learning forecasting paradigms.

Curated by Faculty: S.Bosu Babu
Department of CSE (Data Science)
Mohan Babu University

Chapter 1: Introduction to Time Series Data & Fundamentals

1. Concept and Theoretical Foundation:
Introduction to Time Series Data & Fundamentals represents a crucial pillar in modern temporal data science. Temporal ordering imposes specific dependency structures that standard independent and identically distributed (i.i.d.) assumptions cannot accommodate. Understanding the underlying mathematical formulation and algorithmic behavior is essential for high-fidelity predictive modeling.

2. Methodological Formulation & Key Principles:
When analyzing temporal series, practitioners must decompose signals into long-term trends, periodic seasonal patterns, cyclic variations, and irregular stochastic residual noise. Rigorous statistical testing ensures model assumptions are verified prior to parameter estimation.

3. Practical Implementation with Python:
In Python, tools such as statsmodels, pandas, and scikit-learn provide standardized APIs for model fitting. Data preprocessing requires continuous timestamp indexing, frequency inference, missing value interpolation, and forward/backward imputation methods.

4. Case Studies and Laboratory Exercises:
Practical applications include financial market volatility prediction, electrical load forecasting, network traffic telemetry, and supply chain inventory planning. Students are expected to evaluate residual diagnostics using Q-Q plots, Ljung-Box tests, and out-of-sample backtesting validation.

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Chapter 2: Stationarity, Unit Root Tests & ADF Methodology

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Chapter 3: Autocorrelation (ACF) and Partial Autocorrelation (PACF)

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Chapter 4: Autoregressive (AR) and Moving Average (MA) Processes

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Chapter 5: ARIMA and Seasonal ARIMA (SARIMA) Architectures

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Chapter 6: Exponential Smoothing & Holt-Winters Forecasting

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Chapter 7: Feature Engineering for Temporal Data (Lags & Rolling Statistics)

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Chapter 8: Machine Learning for Forecasting (XGBoost & Prophet)

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Machine Learning for Forecasting (XGBoost & Prophet) represents a crucial pillar in modern temporal data science. Temporal ordering imposes specific dependency structures that standard independent and identically distributed (i.i.d.) assumptions cannot accommodate. Understanding the underlying mathematical formulation and algorithmic behavior is essential for high-fidelity predictive modeling.

2. Methodological Formulation & Key Principles:
When analyzing temporal series, practitioners must decompose signals into long-term trends, periodic seasonal patterns, cyclic variations, and irregular stochastic residual noise. Rigorous statistical testing ensures model assumptions are verified prior to parameter estimation.

3. Practical Implementation with Python:
In Python, tools such as statsmodels, pandas, and scikit-learn provide standardized APIs for model fitting. Data preprocessing requires continuous timestamp indexing, frequency inference, missing value interpolation, and forward/backward imputation methods.

4. Case Studies and Laboratory Exercises:
Practical applications include financial market volatility prediction, electrical load forecasting, network traffic telemetry, and supply chain inventory planning. Students are expected to evaluate residual diagnostics using Q-Q plots, Ljung-Box tests, and out-of-sample backtesting validation.

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Chapter 9: Deep Learning for Time Series (RNN, LSTM & GRU)

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Chapter 10: Model Validation, Forecast Metrics (MAE/RMSE/MAPE) & Deployment

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Model Validation, Forecast Metrics (MAE/RMSE/MAPE) & Deployment represents a crucial pillar in modern temporal data science. Temporal ordering imposes specific dependency structures that standard independent and identically distributed (i.i.d.) assumptions cannot accommodate. Understanding the underlying mathematical formulation and algorithmic behavior is essential for high-fidelity predictive modeling.

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